

Nonlinear Uncertainty Propagation for Perturbed Two-Body Orbits

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The main objective of this paper is to present the development of the computational methodology, based on the Gaussian mixture model, that enables accurate propagation of the probability density function through the mathematical models for orbit propagation. The key idea is to approximate the density function associated with orbit states by a sum of Gaussian kernels. The unscented transformation is used to propagate each Gaussian kernel locally through nonlinear orbit dynamical models. Furthermore, a convex optimization problem is formulated by forcing the Gaussian mixture model approximation to satisfy the Kolmogorov equation at every time instant to solve for the amplitudes of Gaussian kernels. Finally, a Bayesian framework is used on the Gaussian mixture model to assimilate observational data with model forecasts. This methodology effectively decouples a large uncertainty propagation problem into many small problems. A major advantage of the proposed approach is that it does not require the knowledge of system dynamics and the measurement model explicitly. The simulation results are presented to illustrate the effectiveness of the proposed ideas.

I. Introduction

SPACE situational awareness involves collecting resident space objects (RSOs) information, usually orbital position, and the assessment of the effect of this information on future state information in relation to other RSOs. A common example involves assigning the probability of collision of an object with another object. Commercial and military analysts routinely use both observation data and forward orbit propagation models to assist space surveillance. The U.S. Air Force collects information for the purpose of space surveillance through a global network of radars and optical sensors to maintain a catalog of over 20,000 RSOs, both active and inactive. However, because of the large number of RSOs and the limited number of sensors available to track these objects, it is impossible to maintain persistent surveillance on all objects, and therefore there is inherent uncertainty and latency in the catalog.

The fusion of observational data with orbit state models promises to provide a greater understanding of the physical phenomenon than either approach alone can achieve. The most critical challenge here is to provide a quantitative assessment of how closely our estimates reflect reality in the presence of model uncertainty, discretization errors (and their propagation to the final predictions), and measurement errors and uncertainty. The quantitative understanding of uncertainty is essential when predictions are to be used to inform policy making or mitigation solutions in which significant resources are at stake. Model/data fusion had its birth with the development of the Kalman filter (KF) [1]. The KF formulation provides an efficient and optimal tool to estimate the states of a linear system in the presence of noisy measurements and forward model errors. The most popular

extension of the KF for a nonlinear system is known as the extended Kalman filter (EKF) [2]. The EKF process involves the linearization of both system dynamics and the measurement model about the mean, and the state probability density function (PDF) is assumed to be Gaussian. The EKF has been used successfully in many applications; however, it may result in poor performance due to large errors in state estimates or sparse measurement data [3]. Moreover, the linearization requires the calculation of Jacobian of the system equation, which, for a high-dimensional and complex nonlinear system, becomes a formidable task.

The unscented Kalman filter (UKF) [4] is one of the approaches that can be used to overcome some of the deficiencies associated with the EKF. The UKF performs the estimation process by making use of a deterministic sampling technique known as the unscented transformation (UT). A set of deterministic points called sigma points having mean and covariance equal to that of the probability distribution of the given nonlinear system are propagated using the nonlinear function to generate transformed points. The statistics of the new points can be used to estimate the original mean and covariance. This process results in a filter that estimates the mean and covariance more accurately than the EKF. The computational burden involved due to the introduction of sigma points is offset by avoiding the calculation of the system model Jacobian.

Although both the EKF- and UKF-based filters are widely used for state estimation problems, both methods are based upon the very restrictive assumption of Gaussian error. In [5,6], judicious choices for coordinate systems are advocated to decrease the uncertainty propagation errors while making use of linear error theory. Clearly, the Gaussian assumption can work well for moderately nonlinear systems, but it might not be appropriate for certain problems based upon the physical model. For example, the Gaussian assumption for eccentricity or the semimajor axis is completely erroneous. This suggests the need for filters that can incorporate the knowledge about non-Gaussian uncertainty. Various researchers have endeavored to exploit knowledge of statistics, dynamical systems, and numerical analysis to develop nonlinear filtering techniques [7–12] that cater to the various classes of state and parameter estimation problems. The algorithms based on the particle filter have been gaining increasing attention due to recent advances in Monte Carlo sampling methods [11,13–15]. However, Daum in his seminal work [16] discusses that various factors like the volume of state space, in which a conditional PDF is nonvanishing; stationarity of the problem; analytical structure

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of the problem (e.g., linear dynamics, bilinear dynamics, a unimodal PDF, etc.); effective dimensionality of the problem; and so forth strongly affect the computational complexity and performance of the particle filter [16].

Recently, approaches based on the Gaussian mixture model have garnered specific attention in capturing non-Gaussian state density functions. The main idea of the Gaussian sum filter (GSF) is to approximate a non-Gaussian density by a weighted sum of Gaussian kernels [17–19]. Since a Gaussian kernel can be completely defined by its first two moments, only the mean and covariance of each component need to be propagated to quantify the uncertainties in the states of the system. The EKF is generally used to propagate these two quantities. The weights corresponding to each component are kept constant while propagating the density function through the nonlinear dynamical model; however, the weight of each component is updated when the measurements are available. The literature on Gaussian mixtures is rich in the context of nonlinear filtering [3], a GSF with a more advanced measurement update [10], a mixture of Kalman filters [20], and the Gaussian sum particle filtering [21]. Applications and improvements brought to the GSF are continuing endeavors in fields like target tracking [22–24], computer vision [25,26], and geoscience [27]. Recently, the GSFs have been used to propagate the state uncertainty through the unperturbed two-body problem [28,29]. However, in all of these methods, the weights of the Gaussian mixands are kept constant between two measurements that constrains the accuracy of the GSF algorithm in approximating the forecast PDF when the measurements data are sparse [3].

The lack of adaptive algorithms for weights of the Gaussian mixture are felt to be serious disadvantages of many existing algorithms. For systems in which the frequency of the measurements is low and the estimates based on the conditional PDF are also needed in between observations, the accuracy of the forecast PDF is important. For example, an accurate description of the forecast PDF is very important for an accurate and efficient conjunction analysis. Recently, the adaptive Gaussian mixture model (AGMM) was developed, which incorporates the solution for the Fokker–Planck–Kolmogorov equation (FPKE) for continuous-time dynamical systems or the Chapman–Kolmogorov equation (CKE) for discrete-time dynamical systems to solve for the mixand weights [12,30]. The work in [31] extends this result in accurate uncertainty propagation using Gaussian mixtures for the nonlinear filtering problem. The main focus of this paper is to incorporate the complete evolution of the state PDF between two measurements by solving the FPKE for continuous-time dynamical systems and the CKE for discrete-time dynamical systems in the GSF context for the perturbed two-body problem. The conventional AGMM algorithm makes use of the EKF to propagate the mean and covariance of each Gaussian kernel. In this paper, the use of the UKF algorithm is proposed to propagate the mean and covariance of each Gaussian kernel since computation of the Jacobian can be burdensome for complex orbital dynamic models. As a consequence of this, the proposed AGMM algorithm does not require knowledge of system dynamics explicitly, and it considers the system dynamics model as black box. Furthermore, a detailed computational analysis is presented to quantify the computational effort associated with the presented algorithm. Finally, three different measurement scenarios are discussed to show the characterization of non-Gaussian errors in orbit determination.

The structure of the paper is as follows. A brief introduction to the perturbed two-body problem is presented, followed by the development of the AGMM to solve the FPKE and CKE between two measurement time intervals. Next, the computational complexity of the proposed algorithms is discussed. Finally, results for numerical simulation in the discrete-time domain as well as in the continuous-time domain are presented.

II. Problem Statement

The two-body problem is a classical problem in celestial mechanics describing the motion of two heavenly bodies. The two bodies can be two planets or a planet and its natural or artificial moon. In the current context, the motion of a satellite in an orbit around the

Earth is studied. The equations of motion of a satellite while taking into consideration the atmospheric drag effects and the nonspherical shape of the Earth can be expressed as

$$\begin{aligned}\ddot{x} + \frac{\mu x}{r^3} &= J_{2x} + a_{D_x}, & J_{2x} &= -1.5J \left(\frac{\mu}{r^2} \right) \left(\frac{R_e}{r} \right)^2 \left(1 - 5 \frac{z^2}{r^2} \right) \frac{x}{r} \\ \ddot{y} + \frac{\mu y}{r^3} &= J_{2y} + a_{D_y}, & J_{2y} &= -1.5J \left(\frac{\mu}{r^2} \right) \left(\frac{R_e}{r} \right)^2 \left(1 - 5 \frac{z^2}{r^2} \right) \frac{y}{r} \\ \ddot{z} + \frac{\mu z}{r^3} &= J_{2z} + a_{D_z}, & J_{2z} &= -1.5J \left(\frac{\mu}{r^2} \right) \left(\frac{R_e}{r} \right)^2 \left(3 - 5 \frac{z^2}{r^2} \right) \frac{z}{r}\end{aligned}\quad (1)$$

where x , y , and z are the Cartesian coordinates of the satellite and r is the distance of the satellite from the center of the Earth. The gravitational constant is μ , and R_e is the radius of the Earth. J_{2x} , J_{2y} , and J_{2z} indicate the J_2 perturbation effect on the satellite in the x , y , and z directions, respectively, and J is a constant for which the value is 0.001082616. The J_2 perturbation corresponds to the first harmonic of the Earth's gravitational field by assuming the Earth to be a nonspherical body. The atmospheric drag forces, denoted by a_{D_x} , a_{D_y} , and a_{D_z} , are modeled as

$$\begin{aligned}a_{D_x} &= -\frac{1}{2}B\rho(\dot{x} + \Omega_e y) \sqrt{(\dot{x} + \Omega_e y)^2 + (\dot{y} - \Omega_e x)^2 + (\dot{z})^2} \\ a_{D_y} &= -\frac{1}{2}B\rho(\dot{y} - \Omega_e x) \sqrt{(\dot{x} + \Omega_e y)^2 + (\dot{y} - \Omega_e x)^2 + (\dot{z})^2} \\ a_{D_z} &= -\frac{1}{2}B\rho\dot{z} \sqrt{(\dot{x} + \Omega_e y)^2 + (\dot{y} - \Omega_e x)^2 + (\dot{z})^2}\end{aligned}\quad (2)$$

where B is the ballistic coefficient, Ω_e is the rotational speed of the Earth in radians per second, and ρ is the atmospheric density. The exponential model ([32] Table 8.4) is used in this paper for modeling atmospheric density,

$$\rho = \rho_0 \exp(-(R - R_e) - h_0)/H); \quad R = \sqrt{x^2 + y^2 + z^2} \quad (3)$$

where ρ_0 , h_0 , and H are the reference density, reference altitude, and scale height, respectively. A detailed discussion of the modeling of the atmospheric drag and the calculation of the drag coefficient is in [32]. Perfect knowledge of the system dynamics is assumed; i.e., there is no process noise. However, there are uncertainties in the initial states that need to be propagated to quantify the uncertainties creeping in the states over time.

III. Uncertainty Characterization

In the standard orbit dynamical model, the orbit state variables are assumed to be deterministic quantities. A key idea of this work is to replace the time evolution of the orbit state vector $\mathbf{x}(t)$ by the time evolution of the PDF $p(\mathbf{x}(t))$. Thus, the position and velocity of the RSO are assumed to be a random vector $\mathbf{x}(t)$, for which the time evolution is given by the following stochastic differential equation:

$$\dot{\mathbf{x}}(t) = \mathbf{f}(t, \mathbf{x}(t)) + \mathbf{g}(t, \mathbf{x}(t))\eta(t), \mathbf{x}(t_0) = \bar{\mathbf{x}}_0 \quad (4)$$

The random function $\eta(t)$ represents stochastic forcing term that might be added to account for any modeling error. These random functions may be modeled as a Gaussian process with the specified correlation function $\mathbf{Q}(t)$. The discrete-time counterpart of Eq. (4) is

$$\mathbf{x}_{k+1} = \boldsymbol{\phi}(k, \mathbf{x}_k) + \boldsymbol{\eta}_{k+1}, \quad \mathbf{x}_0 = \bar{\mathbf{x}}_0 \quad (5)$$

where $\boldsymbol{\phi}(k, \mathbf{x}_k)$ can be thought of as the flow of nonlinear system described by Eq. (4). It should be noticed that $\boldsymbol{\phi}(k, \mathbf{x}_k)$ can be obtained by a numerical solution of Eq. (4).

The exact time evolution of the state PDF is given by the FPKE [33]:

$$\begin{aligned} \frac{\partial}{\partial t} p(\mathbf{x}(t)) &= \mathcal{L}_{FP} p(\mathbf{x}(t)) = - \sum_{i=1}^n \frac{\partial}{\partial \mathbf{x}_i} \mathbf{f}(t, \mathbf{x}) p(\mathbf{x}(t)) \\ &+ \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \frac{\partial^2}{\partial \mathbf{x}_i \partial \mathbf{x}_j} \mathbf{g}(t, \mathbf{x}) \mathbf{Q} \mathbf{g}^T(t, \mathbf{x}) p(\mathbf{x}(t)) \end{aligned} \quad (6)$$

In the discrete-time domain, one needs to solve the CKE [33] to obtain the PDF at time $k + 1$:

$$p(t_{k+1}, \mathbf{x}_{k+1}) = \int p(t_{k+1}, \mathbf{x}_{k+1} | t_k, \mathbf{x}_k) p(t_k, \mathbf{x}_k) d\mathbf{x}_k \quad (7)$$

The FPKE/CKE is a formidable equation to solve because of 1) positivity of the PDF; 2) the normalization constraint of the PDF, $\int_{\mathbb{R}^n} p(\mathbf{x}(t)) d\mathbf{x} = 1$; and 3) no fixed solution domain (how to impose boundary conditions in a finite region and restrict numerical computation to regions in which $p > \sim 10^{-9}$).

Analytical solutions for the FPKE/CKE exist for a certain class of dynamical systems and mainly for stationary density functions [33,34]. Thus, researchers are actively looking at numerical approximations to solve the FPKE/CKE [35–40], generally using the variational formulation of the problem. These traditional numerical approaches that discretize the space in which the PDF lies suffer from the curse of dimensionality. To overcome this obstacle, the recently developed AGMM to accurately solve the FPKE/CKE associated with the dynamical model is used in this work.

A. Adaptive Gaussian Mixture Model

The key idea of the AGMM is to approximate the state PDF by a finite sum of Gaussian kernels, $\mathcal{N}(\mathbf{x}_{k+1} | \mu_{k+1}^i, P_{k+1}^i)$, with mean μ_{k+1}^i and covariance P_{k+1}^i ,

$$\hat{p}(t_k, \mathbf{x}_k) = \sum_{i=1}^N w_k^i \mathcal{N}(\mathbf{x}_k | \mu_k^i, P_k^i) \quad (8)$$

where w_i denotes the amplitude of i th Gaussian kernel in the mixture. The positivity and normalization constraints on $\hat{p}(t, \mathbf{x})$ lead to weights being nonnegative and the summation of weights being 1. Since each Gaussian kernel is locally capturing the non-Gaussian PDF, one can use linear system theory to propagate the mean and covariance of each Gaussian kernel. In [31], the efficacy of using the Jacobian of the system dynamics is demonstrated to propagate the mean and covariance of each Gaussian kernel. The requirement of the linearized model and the calculation of the Jacobian involved in the propagation of the mean and covariance of each Gaussian kernel can be replaced by making use of the UT [4]. The application of the UT to propagate the first two moments of each Gaussian kernel does not require the computation of any Jacobian matrix, making it attractive when the system dynamics or flow is not known explicitly.

To determine the unknown weights, the CKE or FPKE error is used as a feedback to update the weights of different Gaussian components in the mixture PDF as described in [31]. It should be noted that the formulation involving the CKE is derivative free if one makes use of the UT to propagate the mean and covariance of each Gaussian component. The minimization of the CKE error leads to the solution of the quadratic convex optimization problem [31],

$$\begin{aligned} \min_{\mathbf{w}_{k+1}} J &= \frac{1}{2} \mathbf{w}_{k+1}^T M \mathbf{w}_{k+1} - \mathbf{w}_{k+1}^T N \mathbf{w}_k, \\ \text{such that } 1^T \mathbf{w}_{k+1} &= 1, \quad \mathbf{w}_{k+1} \geq \mathbf{0} \end{aligned}$$

where $\mathbf{w}_{k+1} = [w_{k+1}^1, w_{k+1}^2, \dots, w_{k+1}^N]^T$ and the components of matrix M and the vector $\mathbf{y} \in \mathbb{R}^{N \times 1}$ are given as

$$\begin{aligned} m_{ij} &= |2\pi(P_{k+1}^i + P_{k+1}^j)|^{-1/2} \\ &\times \exp \left[-\frac{1}{2} (\mu_{k+1}^i - \mu_{k+1}^j)^T (P_{k+1}^i + P_{k+1}^j)^{-1} (\mu_{k+1}^i - \mu_{k+1}^j) \right] \end{aligned} \quad (9a)$$

$$\begin{aligned} y_i &= \sum_{j=1}^N w_k^j n_{ij}, \quad n_{ij} = \int \mathcal{N}(\phi(k, \mathbf{x}_k) | \mu_{k+1}^i, \\ &P_{k+1}^i + \mathbf{Q}_k) \mathcal{N}(\mathbf{x}_k | \mu_k^j, P_k^j) d\mathbf{x}_k \end{aligned} \quad (9b)$$

A similar optimization problem can be posed for the weight update in the continuous-time domain by solving the FPKE as explained in [12,31]. This methodology effectively decouples a large uncertainty propagation problem into many small problems. As a consequence, the solution algorithm can be parallelized on most high-performance computing systems. More details of the AGMM formulation and comparison with several conventional methods, using several benchmark problems, can be found in [12,30,31,41–43].

B. Numerical Integration

One needs to solve for m_{ij} and n_{ij} before solving the optimization problem. While m_{ij} can be computed analytically, one needs to resort to numerical methods to find the integral involved in n_{ij} . The accurate evaluation of these integrals constitutes an important part of the computation load associated with the AGMM methodology. Ideally, one can use the Gauss–Hermite (GH) quadrature rule to evaluate these integral with

$$\begin{aligned} n_{ij} &= \int \underbrace{\mathcal{N}(\phi(k, \mathbf{x}_k) | \mu_{k+1}^i, P_{k+1}^i + \mathbf{Q}_k)}_{g(\mathbf{x}_k)} \mathcal{N}(\mathbf{x}_k | \mu_k^j, P_k^j) d\mathbf{x}_k \\ &\approx \sum_{i=0}^{n_p} w^{(i)} g(\mathbf{x}_k^{(i)}) \end{aligned} \quad (10)$$

where n_p is the total number of points used in the approximation of n_{ij} . For one-dimensional integrals, one needs m quadrature points according to the GH quadrature scheme to exactly reproduce the expectation integrals of polynomials with degree $2m - 1$ or less [44]. However, in the generic n dimension, one needs to take the tensor product of one-dimensional quadrature points, which would yield a total of $n_p = m^n$ quadrature points. This is a nontrivial number of points that might make the calculation of the integral computationally expensive.

The capability of the unscented transformation to approximate the first two moments of a Gaussian density function provides a strong motivation to use sigma points in the approximation of the integrals mentioned in Eq. (10). As mentioned earlier, $2n + 1$ sigma points can be used to approximate the first three moments of an n -dimensional Gaussian random vector. For the same random vector, the GH rule needs $n_p = m^n$ points to exactly match the first $2m - 1$ moments. Using $m = 2$, the GH rule needs $n_p = 2^n$ points to exactly match the first three moments. For a six-dimensional system, 13 sigma points can be used for the approximation compared to 64 quadrature points as in the GH rule. There are many versions of the unscented transformation to approximate the expected value of a random function. Other than $2n + 1$ sigma points, one can use $n + 2$ simplex unscented points [45] or $6n + 1$ sigma points [46] to approximate the integral (10). A comparison of these methods is provided later to show their computational efficiency and accuracy.

IV. Computational Complexity

The time elapsed in completing one iteration using the AGMM, in general, depends upon the state dimension and the number of Gaussian kernels in the mixture model. The computational complexity is mainly dictated by the number of operations required to propagate the mean and covariance of each Gaussian kernel and the calculation of N and M matrices. The prediction step in the UKF involves the calculation of matrix square root to find sigma points. For an n -dimensional system, one needs to find the square root of the $n \times n$ matrix, which has a computational complexity of n^3 [47]. The complexity in the calculation of matrix M is $O(N^2 n^3)$, where N is the number of Gaussian components. The computational complexity in the calculation of N is $O(Nn^3 + N^2 L)$, where L is the number of sigma points. The convex optimization process is of the order of $O(SN^5)$ [48], where S is the length of input to the optimization

problem. The total time elapsed in every prediction stage is

$$O(n^3 + N^2n^3 + N^2L + SN^5) \quad (11)$$

For a large number of Gaussian components, serial computation of the N matrix, itself, may take a long time on a normal computer. It leads to the need for parallelization of the code.

The code for the implementation of the AGMM algorithm was written in C language in the Linux platform. There are two classical methods of parallelization, using OpenMP and using MPI in the Linux platform. Each of these methods is best suited for different purposes. While OpenMP is best suited to loop-level parallelization, MPI is the best choice for task-level parallelization. However, there is one inherent disadvantage of OpenMP, as it fails to parallelize a loop in which operations depend upon the previous iteration. Parallelization of a code, on the other hand, requires identification of the operations that are computationally most expensive. Any effort to parallelize all the loops or jobs may lead to worse performance, as a large amount of time may be spent in the synchronization of threads as in OpenMP or processors as in MPI. For the AGMM code, there are three different operations that are computationally expensive: 1) the prediction stage in the Kalman filter for each of the Gaussian components, 2) the calculation of the $M_{N \times N}$ matrix, and 3) the calculation of the $N_{N \times N}$ matrix.

For orbital dynamics problem, the calculation of the N matrix is computationally the most expensive of the three since it involves numerical integration in six-dimensional space. So, only the code that computes the N matrix is parallelized in this work.

Since the calculation of this matrix depends upon the values of the previous iteration, MPI is used to parallelize the code. Since MPI cannot parallelize a nested loop, only the outermost loop of the code is parallelized. Each processor computes the values of all the columns in the rows assigned to them.

Each processor begins with a row equal to its rank. Thereafter, the distribution of the rows to each processor is done so that each of the processors updates different rows. The following scheme is used:

$$\begin{aligned} \text{for } (i = \text{ran } k; i < \text{number of Gaussian components;} \\ i = i + \text{number of processors}) \end{aligned}$$

Thus, each processor updates different rows. At the end of the calculation of rows by each processor, the values are sent to the master processor, which, in the given case, was the processor with rank 0. This leads to a reliable and fast method of computation. Parallelization can be further improved by combining OpenMP and MPI. There are some operations in the computation of the N matrix that do not depend upon the previous iterations and can be effectively parallelized by OpenMP.

V. Numerical Simulation

A satellite in low Earth orbit at an altitude of 622 km is considered for simulation purposes. Both the atmospheric drag and J_2 perturbation are considered in the orbit dynamical model as discussed in Sec. II. The parameters pertaining to the drag model are assumed to be $\rho_0 = 1.454 \times 10^{-13} \text{ kg} \cdot \text{m}^{-3}$, $h_0 = 600 \text{ km}$ and $H = 71.835 \text{ km}$ for this work. The initial state PDF of the system is assumed to be Gaussian with the following mean and covariance:

$$\mu_0 = \begin{bmatrix} \underbrace{7 \times 10^3 \quad 0 \quad 0}_{\text{km}} & \underbrace{0 \quad -1.0374 \quad 7.4771}_{\text{km/s}} \end{bmatrix}$$

$$P_0 = \text{diag} \left(\underbrace{\begin{pmatrix} 0.01 & 0.01 & 0.01 \end{pmatrix}}_{\text{km}^2}, \underbrace{\begin{pmatrix} 0.000001 & 0.000001 & 0.000001 \end{pmatrix}}_{\text{km}^2/\text{s}^2} \right)$$

The covariance matrix P_0 reflects a standard deviation of 100 m in the initial estimate of the three position coordinates and a standard deviation of 1 m/s in the estimate of the velocity. The time period of the satellite is 0.8104 hours.

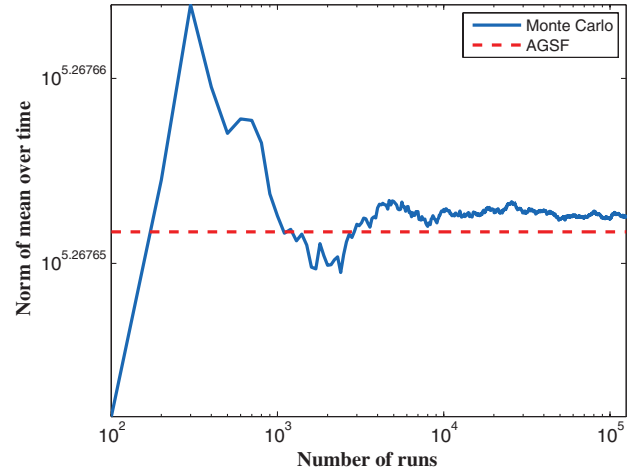


Fig. 1 Convergence of Monte Carlo for different runs.

The initial state PDF is propagated through the orbit dynamics for 6 hours using the AGMM and Monte Carlo (MC) methods with 125,000 runs. The convergence of the norm of the mean over time for different numbers of MC runs along with the norm of the mean computed through the AGMM is shown in Fig. 1. From this graph, it is clear that the mean computed using the AGMM is accurate within the convergence error of MC runs.

In addition to the initial Gaussian PDF, 48 mixture components with zero weight are introduced along the principal axis of the initial covariance matrix. The unit weight given to the original component and zero to other components ensures that the initial PDF of the Gaussian mixture is same as the original PDF. The AGMM updates these weights by minimizing the error between the actual PDF given by the FPKE/CKE and the approximated PDF calculated using the Gaussian mixture. The weights are updated 100 times in an orbit. The N matrix as in Eq. (9b) is of size 49×49 , and one needs to use efficient numerical integration to reduce the computational burden. Table 1 shows the efficacy of different methods of numerical integration while calculating the N matrix. The table shows the Frobenius norm of the matrix using different integration schemes. One can see that the simplex points give an accurate result with a lower number of points and hence have less computational complexity.

A. Results

Contours corresponding to the prior PDF, while taking into consideration the J_2 and drag effects, are shown in Figs. 2 and 3. The distribution changes from Gaussian at $t = 0$ to non-Gaussian after some time. Since no measurements are available, a weight update cannot be made in the UKF, and, as such, its contours are not able to provide an efficient quantification of uncertainty as the distribution changes to non-Gaussian. AGMM contours, on the other hand, are able to capture the uncertainty in an efficient manner and are able to match the behavior of MC runs.

Figure 4 shows the skewness in the states. The amount of skewness indicates the non-Gaussian nature of the PDF, and the negative and positive values of the skewness indicate whether the PDF is skewed toward the left or toward the right, respectively. From these, the AGMM is able to capture the skewness and is able to give a good approximation of the actual PDF.

Table 1 Different numerical integration schemes for the two-body problem

Integration Scheme	Point in each direction	Total points	Frobenius norm of N
Gauss–Hermite	1	1	3.61215e + 009
Gauss–Hermite	2	64	1.164975e + 008
Gauss–Hermite	3	729	1.164975e + 008
$2n + 1$ Sigma points	— —	13	3.49492e + 008
$6n + 1$ Sigma points	— —	37	1.164975e + 008

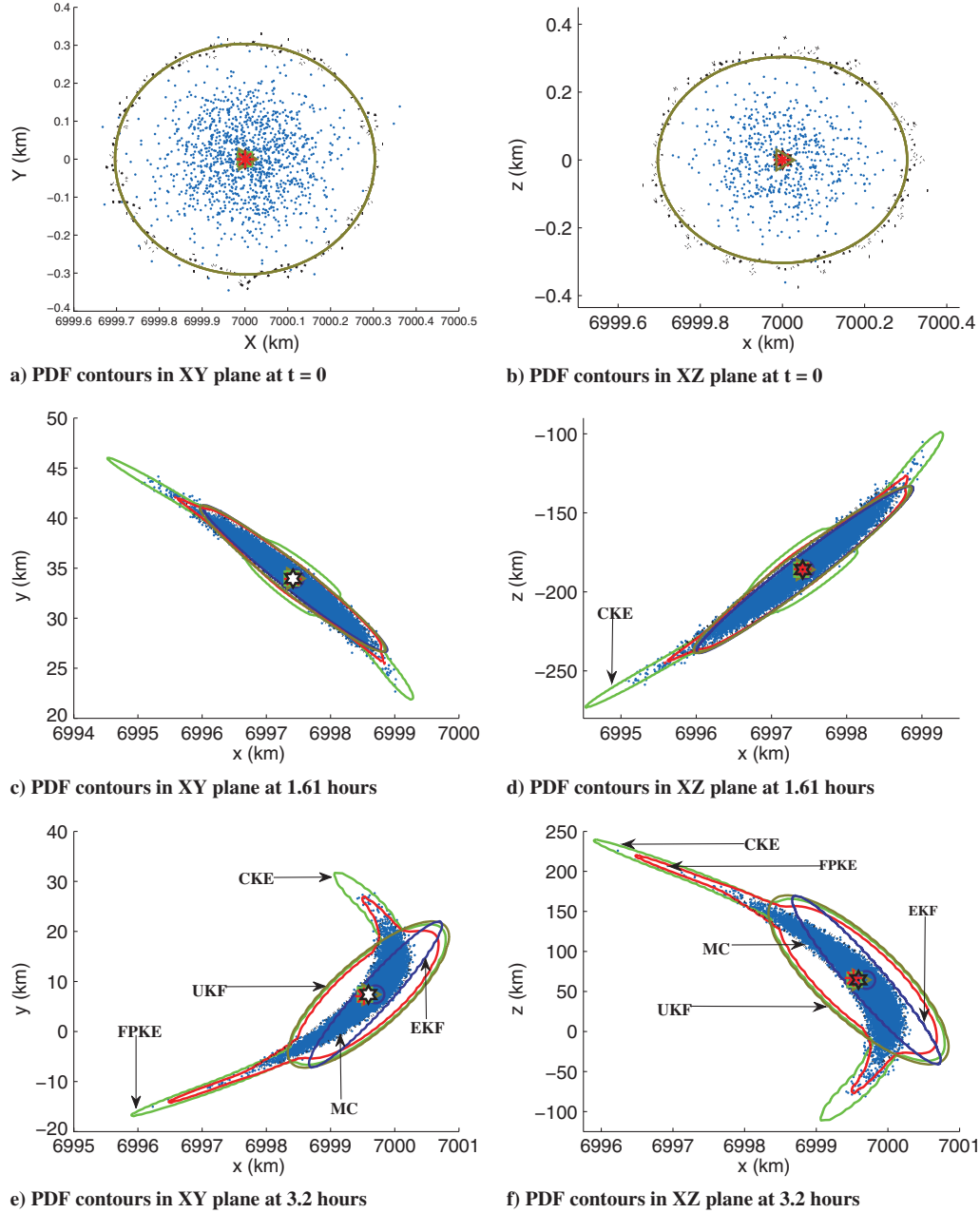


Fig. 2 Contours corresponding to the prior PDF in Cartesian coordinates with drag using the MC, AGSF (Fokker–Plank and Chapman–Kolmogorov), UKF, and EKF at different times.

B. Measurement Update

Measurements are available when the satellite is in the field of view of the radar of an observation center. For simulation purposes, we consider the observation center to be located at 39.007° latitude and 104.883° longitude near the U.S. Air Force Academy in Colorado Springs, Colorado. The Cartesian coordinates of this location are given as

$$\mathbf{r}_{\text{Site}} = [-1275.1219 \quad -4797.9890 \quad 3994.2975] \text{ km}$$

The location of this radar is used for the measurement update. It is assumed that the measurements are available after 5.6585 hours, i.e., the time at which the last time update was made. One can see from Fig. 3 that the PDF contours at this time are non-Gaussian. Six different cases are discussed, four of which assume that the measurements provide the location of the satellite in the tail of the prior contours in the x - y plane, while two of them assume that the measurements provide the location of the satellite at and near the prior mean. Figure 5 shows the six different cases.

It is assumed that the measurements are available at a discrete time. A general measurement model in discrete time, driven by noise, can be expressed as

$$\mathbf{z}_k = \mathbf{h}(t_k, \mathbf{x}_k) + \boldsymbol{\nu}_k \quad (12)$$

where $\boldsymbol{\nu}_k$ represents the measurement noise, which is uncorrelated with zero mean and the covariance of $\mathbf{R}_k$. The mean and covariance of each Gaussian component can be updated whenever a measurement is available. Figure 6 shows the fusion of measurement data with the state PDF obtained during pure propagation. The parameters of the state PDF are updated using the Bayesian framework, and the resulting process is known as the adaptive Gaussian sum filter (AGSF). One can see from the figure that the AGSF is built upon the framework of classical filtering, but it differs from the latter because of the use of the FPKE/CKE to update weights during propagation. In the absence of the FPKE/CKE error feedback, the AGSF reduces to the classical GSF technique. For the purpose of comparison, we have also implemented the EKF as well as the UKF [4] algorithms to assess the performance of the proposed approach. The weight of each

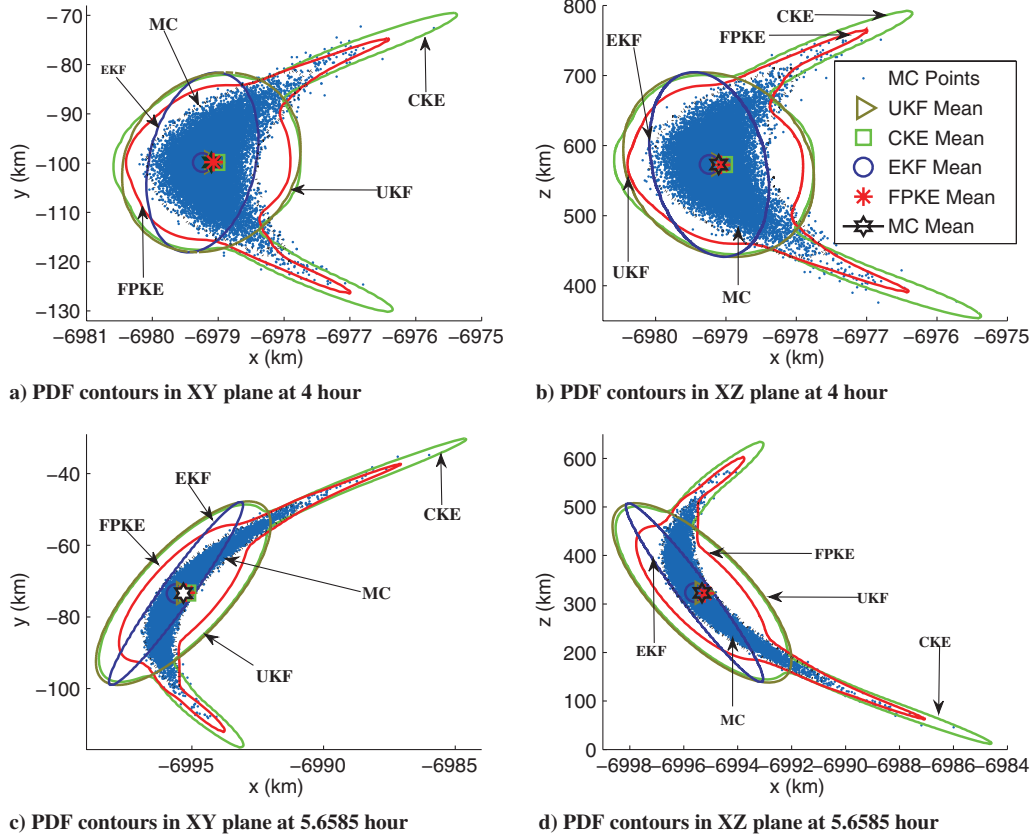


Fig. 3 Contours corresponding to the prior PDF in Cartesian coordinates with drag using the MC, AGSF (Fokker–Plank and Chapman–Kolmogorov), UKF, and EKF at different times.

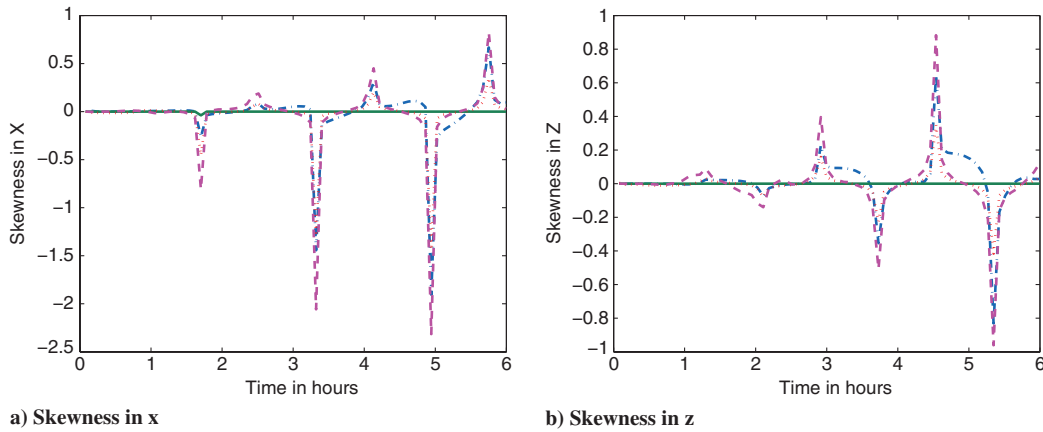


Fig. 4 Skewness in states using the MC, AGMM–FPKE, AGMM–CKE and UKF.

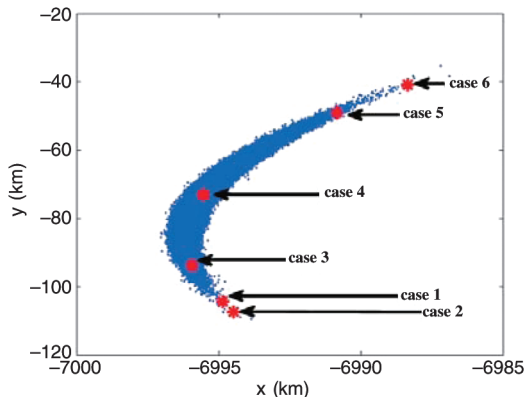


Fig. 5 Different measurement cases. Case 4 corresponds to the prior mean.

Gaussian component is updated using the Bayes rule,

$$\mathbf{w}_{k+1|k+1}^i = \frac{w_{k+1|k}^i \gamma^i}{\sum_{i=1}^N w_{k+1|k}^i \gamma^i} \quad i = 1, 2, \dots, N \quad (13)$$

where N is the total number of Gaussian components and $\gamma^i \sim \mathcal{N}(\mathbf{z}_{k+1} - \boldsymbol{\mu}_{k+1|k}^i, \mathbf{H}_k^{(i)} \mathbf{P}_{k+1|k}^{(i)} (\mathbf{H}_k^{(i)})^T + \mathbf{R}_k)$ for the EKF and $\gamma^i \sim \mathcal{N}(\mathbf{z}_{k+1} - \boldsymbol{\mu}_{k+1|k}^i, \hat{\mathbf{S}}_{k+1}^i)$ for the UKF. Here, $\mathbf{H}_k = \frac{\partial}{\partial \mathbf{x}_k} \mathbf{h}(t_k, \mathbf{x}_k)|_{\mathbf{x}_k = \boldsymbol{\mu}_{k+1|k}^i}$ and $\hat{\mathbf{S}}_{k+1}^i$ is the innovation covariance defined in Ref. [4].

Three different scenarios that correspond to different types of quantities available for measurement are discussed. For each scenario, the six different cases mentioned in Fig. 5 are discussed in the following.

1. Scenario I: Coordinates x , y , and z of Satellite Available for Measurement

The x , y , and z coordinates of the satellite are available for measurement. The measurement model is: $\mathbf{z}_{k+1}(i) = \mathbf{x}_{k+1}(i) + \nu_i$, $i = 1, 2, 3$, where

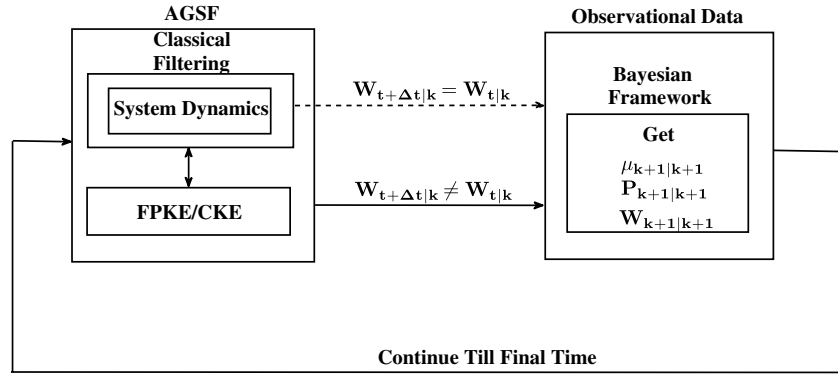


Fig. 6 Fusion of measurement data under the Bayesian framework.

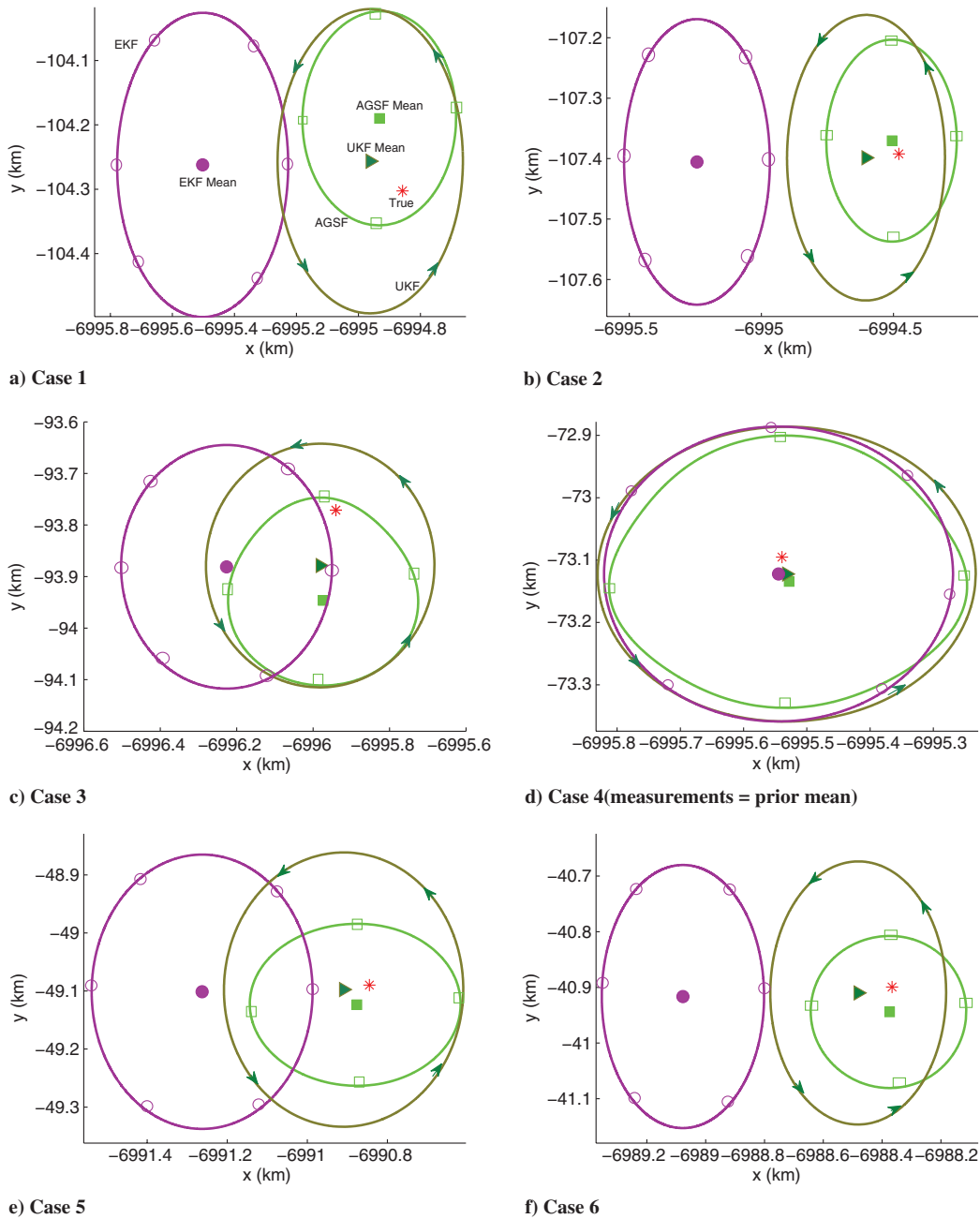


Fig. 7 Contours corresponding to the posterior PDF of the AGSF, UKF, and EKF in Cartesian space when only position measurements are available.

Table 2 Norm of error/variance when only position measurements are available and they are not at the prior mean (pos. error means posterior error, and pos. variance means posterior variance)

Filters	Likelihood	Pos. error in position, km	Pos. variance of position, km ²
AGSF	3.5656e − 004	0.237	0.7837
UKF	7.3787e − 006	0.244	0.0339
EKF	1.674e − 014	1.3329	0.03194

Table 3 Norm of error when range and angular measurements are available and they are not at the prior mean (pos. error means posterior error, and pos. variance means posterior variance)

Filters	Likelihood	Pos. error in position, km	Pos. variance of position, km ²
AGSF	0.006515	2.042	3.4608
UKF	0.006119	7.1679	3.1315
EKF	0.006107	7.7479	0.3987

$$\nu = [\nu_1 \ \nu_2 \ \nu_3]^T \sim \mathcal{N}(\mathbf{0}, \mathbf{R}_1) \text{ and,}$$

$$\mathbf{R}_1 = \text{diag}(\underbrace{0.01 \ 0.01 \ 0.01}_{\text{km}^2})$$

Figure 7 shows the contours at the 99% level corresponding to the posterior PDF for different filters in the Cartesian space when only the x , y , and z coordinates of the satellite are available for

measurement. Table 2 shows the norm of error for different cases when only the position measurements are available. It should be noted that the EKF estimates are statistically inconsistent. The posterior variance computed through the EKF is very small, although the estimation error is very large. This is completely erroneous since one often uses filter variance to represent the confidence in state estimates. One can see that the AGSF consistently outperforms other conventional filters.

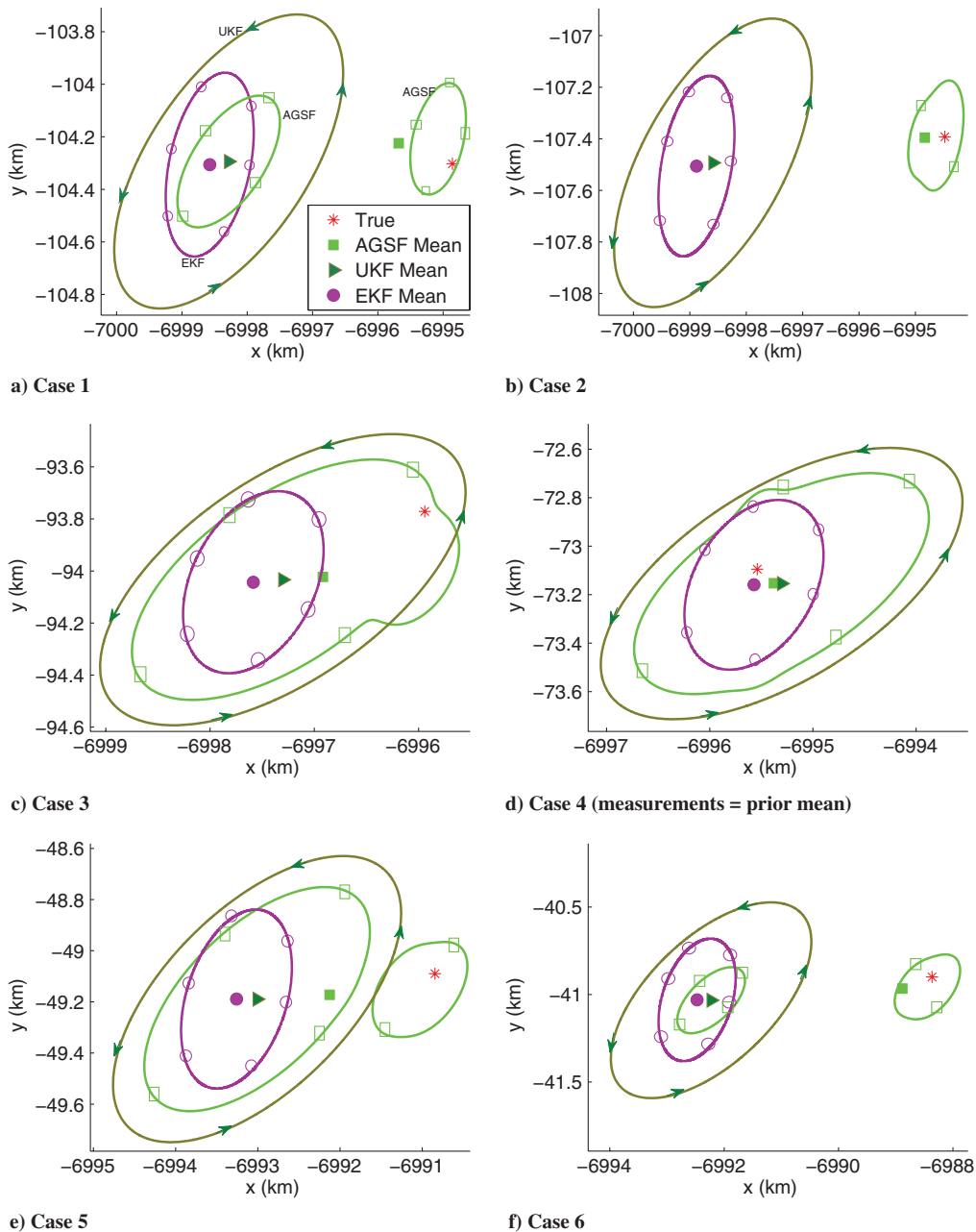


Fig. 8 Contours corresponding to the posterior PDF of the AGSF, UKF, and EKF in Cartesian space when range and angle measurements are available.

2. Scenario II: Range, Azimuthal, and Elevation Angles Available for Measurement

In this scenario, the range, azimuthal, and elevation angles are assumed to be available as measurements:

$$\begin{aligned} \mathbf{z}_{k+1}(1) &= \sqrt{(\mathbf{x}_{k+1}(1) - \text{rSite}(1))^2 + (\mathbf{x}_{k+1}(2) - \text{rSite}(2))^2 + (\mathbf{x}_{k+1}(3) - \text{rSite}(3))^2} + \nu_1 \\ \mathbf{z}_{k+1}(2) &= \tan^{-1} \frac{\mathbf{x}_{k+1}(3) - \text{rSite}(3)}{\sqrt{(\mathbf{x}_{k+1}(1) - \text{rSite}(1))^2 + (\mathbf{x}_{k+1}(2) - \text{rSite}(2))^2}} + \nu_2 \\ \mathbf{z}_{k+1}(3) &= \tan^{-1} \frac{\mathbf{x}_{k+1}(2) - \text{rSite}(2)}{\mathbf{x}_{k+1}(1) - \text{rSite}(1)} + \nu_3 \end{aligned}$$

where,

$$\boldsymbol{\nu} = [\nu_1 \ \nu_2 \ \nu_3]^T \sim \mathcal{N}(\mathbf{0}, \mathbf{R}_2) \text{ and } \mathbf{R}_2 = \text{diag}\left(\underbrace{0.01}_{\text{km}^2} \underbrace{0.0174 \ 0.0174}_{\text{rad}^2}\right)$$

The measurement model for this scenario, i.e., when the range, azimuthal, and elevation angles are available for measurement, is nonlinear in nature. The EKF uses linearization to make the update and, as such, underestimates the covariance. Figure 8 shows contours

at the 99% level for the the posterior PDF in the xy space when the range and angular measurements are available. The AGSF mean shows the least error among all the filters and shows multiple peaks, which indicates a high degree of uncertainty in the mean. The contours corresponding to conventional filters, on the other hand, fail

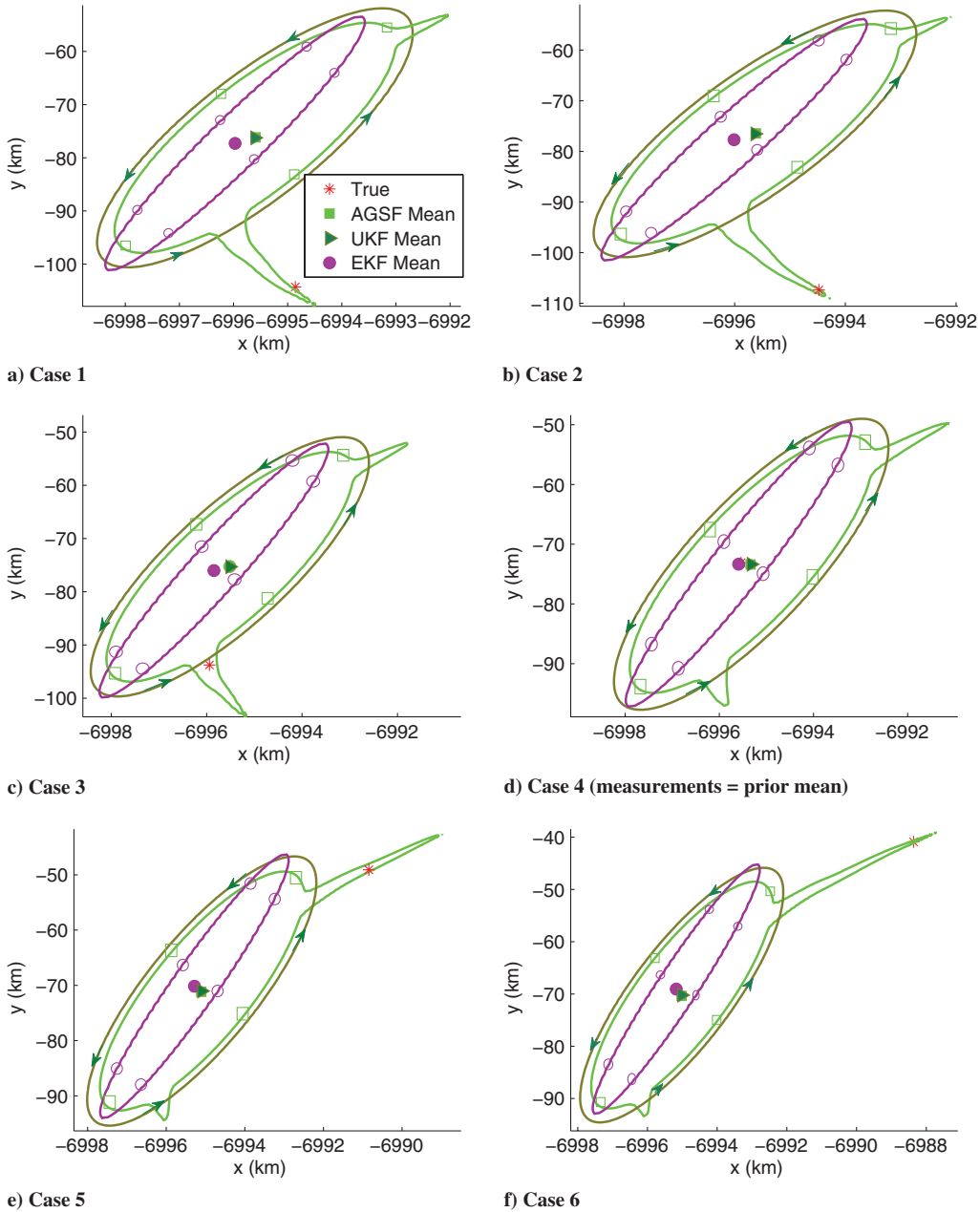


Fig. 9 Contours corresponding to the posterior PDF of the AGSF, UKF, and EKF in Cartesian space when only angle measurements are available.

Table 4 Norm of error when only angular measurements are available and they are not at the prior mean (pos. error means posterior error, and pos. variance means posterior variance)

Filters	Likelihood	Pos. error in position, km	Pos. variance of position, km ²
AGSF	20.0879	469.863	8264
UKF	20.0873	469.944	8439
EKF	20.0873	469.947	8436

to capture even the true mean when the measurement is in the tail region. Table 3 shows the norm of the error for different cases when the range and angular measurements are available.

One can see that the AGSF consistently outperforms other conventional filters. The large value of posterior variance in the case of the AGSF indicates the low confidence in the available measurements. The EKF, in addition to having a large error in the mean, also undermines the posterior variance and thus assumes a high degree of confidence in its mean, which is completely misleading.

3. Scenario III: Azimuthal and Elevation Angles Available for Measurement

Only the azimuthal and elevation angles are available for measurement:

$$\mathbf{z}_{k+1}(1) = \tan^{-1} \frac{\mathbf{x}_{k+1}(3) - \text{rSite}(3)}{\sqrt{(\mathbf{x}_{k+1}(1) - \text{rSite}(1))^2 + (\mathbf{x}_{k+1}(2) - \text{rSite}(2))^2}} + \nu_1$$

$$\mathbf{z}_{k+1}(2) = \tan^{-1} \frac{\mathbf{x}_{k+1}(2) - \text{rSite}(2)}{\mathbf{x}_{k+1}(1) - \text{rSite}(1)} + \nu_2$$

where,

$$\boldsymbol{\nu} = [\nu_1 \ \nu_2]^T \sim \mathcal{N}(\mathbf{0}, \mathbf{R}_3) \quad \text{and} \quad \mathbf{R}_3 = \text{diag} \times \begin{pmatrix} 0.0174 & 0.0174 \\ \text{rad}^2 \end{pmatrix}$$

The PDF of the contours of the azimuthal and elevation angles corresponding to this scenario are plotted. Figure 9 shows the contour corresponding to the posterior PDF of the azimuthal and elevation angles. Similar to the previous two cases, the conventional filters fail to capture the true mean when it is in the tail region. Table 4 shows the norm of the error for different cases in which only the angular measurements are available. As in the previous cases, the AGSF performs better than other filters, although the difference is not remarkable. However, the main advantage of the AGSF methodology is that it is always able to predict the position of the satellite within the 99% contour of the posterior state PDF as shown in Fig. 9.

VI. Conclusions

This paper presents the formulation for accurate uncertainty propagation through orbital mechanics while making use of the differential and integral forms of the Kolmogorov equation. A Gaussian mixture model is used to approximate the state PDF, and, furthermore, a convex optimization problem is posed to update unknown parameters of the mixture model. The proposed methodology effectively decouples a large uncertainty propagation problem into many small problems. As a consequence, the computational complexity can be easily taken care of by using any high-performance computing facility. The proposed methodology is particularly more advantageous than the traditionally used methods based on the Gaussian approximation when system dynamics and the measurement model are highly nonlinear, measurement data are sparse, and one needs to propagate the state uncertainty through the dynamical system for a long duration. This makes the proposed approach very attractive for efficient and accurate conjunction analysis in which one often needs an accurate characterization of the state PDF. In addition to this, the new approach is very attractive for orbital problems

involving high area-to-mass ratio objects under the nonconservative perturbing forces of atmospheric drag and solar radiation pressure. Finally, although the Cartesian coordinates are used to assess the performance of the proposed approach, the developed uncertainty propagation algorithm is generic enough to be used with any orbital dynamical model.

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References

- [1] Kalman, R. E., "A New Approach to Linear Filtering and Prediction Problems 1," *Transactions of the ASME-Journal of Basic Engineering*, Vol. 82, No. Series D, 1960, pp. 35–45.
- [2] Schmidt, S. F., "Application of State-Space Methods to Navigation Problems," *Advanced Control Systems*, Vol. 3, 1966, pp. 293–340.
- [3] Alspach, D., and Sorenson, H., "Nonlinear Bayesian Estimation Using Gaussian Sum Approximations," *IEEE Transactions on Automatic Control*, Vol. 17, No. 4, 1972, pp. 439–448. doi:10.1109/TAC.1972.1100034
- [4] Julier, S., and Uhlmann, J., "Unscented Filtering and Nonlinear Estimation," *Proceedings of the IEEE*, Vol. 92, No. 3, March 2004, pp. 401–422.
- [5] Junkins, J. L., and Singla, P., "How Nonlinear Is It? A Tutorial on Nonlinearity of Orbit and Attitude Dynamics," *Journal of Astronautical Sciences*, Vol. 52, Nos. 1–2, 2004, pp. 7–60.
- [6] Junkins, J. L., Akella, M. R., and Alfrend, K. T., "Non-Gaussian Error Propagation in Orbital Mechanics," *Guidance and Control*, Vol. 44, No. 4, 1996, pp. 283–298.
- [7] Daum, F. E., "A New Nonlinear Filtering Formula for Discrete Time measurements," *24th Inst. of Electrical and Electronics Engineers Conference on Decision and Control*, Vol. 24, Inst. of Electrical and Electronics Engineers, Ft. Lauderdale, FL, 1985, pp. 1957–1958.
- [8] Daum, F. E., "A New Nonlinear Filtering Formula for Non-Gaussian Discrete Time Measurements," *25th Inst. of Electrical and Electronics Engineers Conference on Decision and Control*, Vol. 25, Inst. of Electrical and Electronics Engineers, Athens, Greece, 1986, pp. 1030–1031.
- [9] Daum, F. E., "Exact Finite Dimensional Nonlinear Filters," *24th Inst. of Electrical and Electronics Engineers Conference on Decision and Control*, Vol. 24, Inst. of Electrical and Electronics Engineers, Ft. Lauderdale, FL, 1985, pp. 1938–1945.
- [10] Ito, K., and Xiong, K., "Gaussian Filters for Nonlinear Filtering Problems," *IEEE Transactions on Automatic Control*, Vol. 45, No. 5, 2000, pp. 910–927. doi:10.1109/9.855552
- [11] Sloan, I. H., and Woniakowski, H., "When Are Quasi-Monte Carlo Algorithms Efficient for High Dimensional Integrals?" *Journal of Complexity*, Vol. 14, No. 1, 1998, pp. 1–33. doi:10.1006/jcom.1997.0463
- [12] Terejanu, G., Singla, P., Singh, T., and Scott, P. D., "Uncertainty Propagation for Nonlinear Dynamic Systems Using Gaussian Mixture Models," *Journal of Guidance, Control, and Dynamics*, Vol. 31, No. 6, 2008, pp. 1623–1633. doi:10.2514/1.36247
- [13] Doucet, A., de Freitas, N., and Gordon, N., *Sequential Monte-Carlo Methods in Practice*, Springer-Verlag, Berlin, April 2010, pp. 6–14.
- [14] Khan, Z., Balch, T., and Dellaert, F., "An MCMC-Based Particle Filter for Tracking Multiple Interacting Targets," *Computer Vision-ECCV 2004*, Springer-Verlag, Berlin, 2004, pp. 279–290.
- [15] Khan, Z., Balch, T., and Dellaert, F., "MCMC-Based Particle Filtering for Tracking a Variable Number of Interacting Targets," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, Vol. 27, No. 11, 2005, pp. 1805–1819.
- [16] Daum, F. E., and Huang, J., "Curse of Dimensionality and Particle Filters," *Proceedings of the 2003 Inst. of Electrical and Electronics Engineers Aerospace Conference*, Vol. 4, Inst. of Electrical and Electronics Engineers, Montana, 2003, pp. 1979–1993.
- [17] Lo, J., "Finite-Dimensional Sensor Orbits and Optimal Nonlinear Filtering," *IEEE Transactions on Information Theory*, Vol. 18, No. 5,

- 1972, pp. 583–588.
doi:10.1109/TIT.1972.1054885
- [18] Sorenson, H. W., and Alspach, D., “Recursive Bayesian Estimation Using Gaussian Sums,” *Automatica*, Vol. 7, No. 4, 1971, pp. 465–479.
doi:10.1016/0005-1098(71)90097-5
 - [19] Cameron, A., “Control, and Estimation of Linear Systems with Non-Gaussian A Priori Distributions,” *Proceedings of the 6th Annual Allerton Conference Circuit and System Theory*, IEEE, Urbana, IL, 1968.
 - [20] Chen, R., and Liu, J., “Mixture Kalman Filters,” *Journal of the Royal Statistical Society: Series B*, Vol. 62, No. 3, 2000, pp. 493–508.
doi:10.1111/1467-9868.00246
 - [21] Kotecha, J., and Djuric, P., “Gaussian Sum Particle Filtering,” *IEEE Transactions on Signal Processing*, Vol. 51, No. 10, Oct. 2003, pp. 2602–2612.
doi:10.1109/TSP.2003.816754
 - [22] Blom, H., and Bar-Shalom, Y., “The Interacting Multiple Model Algorithm for Systems with Markovian Switching Coefficients,” *IEEE Transactions on Automatic Control*, Vol. 33, No. 8, 1988, pp. 780–783.
doi:10.1109/9.1299
 - [23] Tam, W. I., Plataniotis, K. N., and Hatzinakos, D., “An Adaptive Gaussian Sum Algorithm for Radar Tracking,” *Signal Processing*, Vol. 77, No. 1, 1999, pp. 85–104.
doi:10.1016/S0165-1684(99)00025-0
 - [24] Musicki, D., Scala, B. L., and Evans, R., “The Integrated Track Splitting Filter—Efficient Multi-Scan Single Target Tracking in Clutter,” *IEEE Transactions on Aerospace and Electronic Systems*, Vol. 43, No. 4, 2007, pp. 1409–1425.
doi:10.1109/TAES.2007.4441748
 - [25] Brasnett, P., Mihaylova, L., Canagarajah, N., and Bull, D., “Particle Filtering with Multiple Cues for Object Tracking in Video Sequences,” *Proceedings of Soc. of Photo-Optical Instrumentation Engineers Annual Symposium*, Vol. 5685, SPIE, San Jose, CA, 2005, pp. 430–441.
 - [26] Han, B., Zhu, Y., Comaniciu, D., and Davis, L., “Kernel-Based Bayesian Filtering for Object Tracking,” *Proceedings of the Inst. of Electrical and Electronics Engineers Conference on Computer Vision and Pattern Recognition*, Inst. of Electrical and Electronics Engineers, San Diego, 2005, pp. 227–234.
 - [27] Santamaria, I., Pantaleon, C., Ibanez, J., and Artes, A., “Deconvolution of Seismic Data Using Adaptive Gaussian Mixtures,” *IEEE Transactions on Geoscience and Remote Sensing*, Vol. 37, No. 2, March 1999, pp. 855–859.
doi:10.1109/36.752203
 - [28] Horwood, J., Aragon, N., and Poore, A., “Gaussian Sum Filters for Space Surveillance: Theory and Simulations,” *Journal of Guidance, Control, and Dynamics*, Vol. 34, No. 6, 2011, pp. 1839–1851.
doi:10.2514/1.53793
 - [29] DeMars, K. J., Bishop, R. H., and Jah, M. K., “Entropy-Based Approach for Uncertainty Propagation of Nonlinear Dynamical Systems,” *Journal of Guidance, Control, and Dynamics*, Vol. 36, No. 4, 2013, pp. 1047–1057.
 - [30] Singla, P., and Singh, T., “A Gaussian Function Network for Uncertainty Propagation Through Nonlinear Dynamical System,” *18th Advances in Astronautical Sciences/AIAA Spaceflight Mechanics Meeting*, AAS Paper 2008-154, 2008.
 - [31] Terejanu, G., Singla, P., Singh, T., and Scott, P. D., “Adaptive Gaussian Sum Filter for Nonlinear Bayesian Estimation,” *IEEE Transactions on Automatic Control*, Vol. 56, No. 9, 2011, pp. 2151–2156.
doi:10.1109/TAC.2011.2141550
 - [32] Vallado, D. A., *Fundamentals of Astrodynamics and Applications*, 3rd ed., Microcosm, 2007, pp. 549–553.
 - [33] Risken, H., *The Fokker-Planck Equation: Methods of Solution and Applications*, Springer-Verlag, Berlin, 1989, pp. 1–12 and 32–62.
 - [34] Fuller, A. T., “Analysis of Nonlinear Stochastic Systems by Means of the Fokker-Planck Equation,” *International Journal of Control*, Vol. 9, No. 6, 1969, pp. 603–655.
doi:10.1080/00207176908905786
 - [35] Lambert, H. C., Daum, F. E., and Weatherwax, J. L., “A Split-Step Solution of the Fokker-Planck Equation for the Conditional Density,” *ACSSC 2006: 40th Asilomar Conference on Signals, Systems and Computers*, Pacific Grove, CA, 2006, pp. 2014–2018.
 - [36] Kumar, M., Singla, P., Chakravorty, S., and Junkins, J. L., “A Multi-Resolution Approach for Steady State Uncertainty Determination in Nonlinear Dynamical Systems,” *38th Southeastern Symposium on System Theory*, IEEE, Cookeville, TN, 2006, pp. 344–348.
 - [37] Kumar, M., Singla, P., Chakravorty, S., and Junkins, J. L., “The Partition of Unity Finite Element Approach to the Stationary Fokker-Planck Equation,” *2006 AIAA/Advances in Astronautical Sciences Astrodynamics Specialist Conference and Exhibit*, AIAA Paper 2006-6285, 2006.
 - [38] Kumar, M., Chakravorty, S., Singla, P., and Junkins, J. L., “The Partition of Unity Finite Element Approach with Hp-Refinement for the Stationary Fokker-Planck Equation,” *Journal of Sound and Vibration*, Vol. 327, Nos. 1–2, 2009, pp. 144–162.
doi:10.1016/j.jsv.2009.05.033
 - [39] Muscolino, G., Ricciardi, G., and Vasta, M., “Stationary and Non-Stationary Probability Density Function for Non-Linear Oscillators,” *International Journal of Non-Linear Mechanics*, Vol. 32, No. 6, 1997, pp. 1051–1064.
doi:10.1016/S0020-7462(96)00134-5
 - [40] Paola, M. D., and Sofi, A., “Approximate Solution of the Fokker-Planck-Kolmogorov Equation,” *Probabilistic Engineering Mechanics*, Vol. 17, No. 4, 2002, pp. 369–384.
doi:10.1016/S0266-8920(02)00034-6
 - [41] Terejanu, G., Singla, P., Singh, T., and Scott, P. D., “A Novel Gaussian Sum Filter Methods for Accurate Solution to Nonlinear Filtering Problem,” *International Conference on Information Fusion*, IEEE, Cologne, Germany, 2008, pp. 1–8.
 - [42] Terejanu, G., Singla, P., Singh, T., and Scott, P. D., “Uncertainty Propagation for Nonlinear Dynamical Systems Using Gaussian Mixture Models,” *Proceedings of the AIAA Guidance, Navigation and Control Conference and Exhibit*, AIAA Paper 2008-7472, 2008.
 - [43] Terejanu, G., George, J., and Singla, P., “An Adaptive Gaussian Sum Filter for The Spacecraft Attitude Estimation Problem,” *Journal of Astronautical Sciences*, Vol. 57, Nos. 1–2, Jan.–June 2009, pp. 31–45.
 - [44] Stroud, A. H., and Secrest, D., *Gaussian Quadrature Formulas*, Prentice-Hall, Upper Saddle River, NJ, 1966, pp. 1–28.
 - [45] Julier, S., and Uhlmann, J., “Reduced Sigma Point Filters for the Propagation of Means and Covariances Through Nonlinear Transformations,” *Proceedings of the 2002 American Control Conference*, Vol. 2, American Automatic Control Council, Anchorage, AK, 2002, pp. 887–892 (Inst. of Electrical and Electronics Engineers Catalog Number CH37301).
 - [46] Wu, Y., Wu, M., Hu, D., and Hu, X., “An Improvement to Unscented Transformation,” *17th Australian Joint Conference on Artificial Intelligence*, Springer, Berlin/Heidelberg/New York, 2004, pp. 1024–1029.
 - [47] Papadimitriou, C. H., *Computational Complexity*, Wiley, Hoboken, NJ, 2003, pp. 366–367.
 - [48] Kozlov, M., Tarasov, S., and Khachiyan, L., “Polynomial Solvability of the Convex Quadratic Programming,” *Doklady Akademiia Nauk SSSR [Soviet Physics — Doklady]*, Vol. 248, 1979, pp. 1049–1051.
doi:10.1016/0041-5553(80)90098-1